



INTERNATIONAL
TELECOMMUNICATION UNION
**TELECOMMUNICATION
STANDARDIZATION SECTOR**
STUDY PERIOD 2022-2024

SG5-xxxxx-xx

STUDY GROUP 5

Original: English

Question(s): 7/5

Sophia Antiopolis, 13 June - 23 June
2023

CONTRIBUTION

Source: India

Title: Proposal for update in baseline text ITU-T L.Suppl.ICIP "Best practices for implementing the circular economy in ICT management"

Contact: Avadhesh Singh Tel: +917011207136
Telecommunication Engineering Centre, Email: avadhesh.singh48@gov.in
Ministry of Communications
India

Contact: Ziaur Rahman Tel: +919910872353
Telecommunication Engineering Centre, Email: ziaur.rahman48@gov.in
Ministry of Communications
India

Abstract: In this contribution, the initial draft for Supplement L.Suppl.ICIP - "Best practices for implementing the circular economy in ICT management has been improved and enriched with content to bridge the gap which includes addition of new use cases for circular practices, detailing under heading of public donation of decommissioned ICT devices, communal involvement, policy mechanism, implementation of the circular economy. Further a new section 'Challenges in Implementation of the circular economy' has also been added in this contribution."

Table of Contents

1	Scope	51
2	References	51
3	Definitions	51
3.1	Terms defined elsewhere	52
3.2	Terms defined in this Supplement.....	52
4	Abbreviations and acronyms	62
5	Conventions	63
6	Commitments to circular ICT management	63
7	Summary of circular ICT management use cases.....	63
7.1	Digitalisation of circular design.....	73
7.2	Green and due diligence for public procurement of sustainable ICT products	73
7.3	Repairability and repair	83
7.4	Circular business models	83
7.5	Reuse and refurbishment	84
7.6	Recycling	84
7.7	Public donation of decommissioned ICT devices.....	94
7.8	Extended producer responsibility.....	94
7.9	Environmental impacts: data transparency, accountability, verifiability	94
7.10	Communal involvement.....	94
7.11	Circular markets	94
8	Catalogue of cases, initiatives and policy mechanisms.....	104
9	Metrics for impact and implementation	114
10	<u>Challenges in implementation</u>	<u>114</u>

Draft Supplement L.Suppl.ICIP to ITU-T L-series of Recommendations

Best practices for implementing the circular economy in ICT management

Summary

This Supplement aims to provide a series of best practices for implementing the circular economy across the ICT value chain. It will contain best practices of initiatives, policy instruments, circular processes, and other examples for improving the circularity of devices and infrastructures targeting both the public and private sectors. The cases will be selected based on the reported or assessed impacts.

Each case will be analysed based on its replicability, scalability, and sustainability impacts from the social, economic, and environmental perspectives. The analysis will be carried out in line with global commitments such as the UN SDGs, and those related to the ICT sector in terms of ITU initiatives such as the Connect-2030 agenda and Recommendations such as the L series, among others. This Supplement will result in a classification of the specific cases and general guidelines, goals, and metrics for the implementation (including design, bootstrap, replication, and evaluation) of scalable circular models in cities, communities, countries, and regions.

This Supplement takes into special consideration draft Recommendation ITU-T L.1033 on guidance for institutions of higher learning. It will contribute to the effective life cycle management of e-equipment and e-waste.

Keywords

<Mandatory>

Introduction

<Optional – This clause should appear only if it contains information different from that in Scope and Summary>

Draft Supplement L.Suppl.ICIP to ITU-T L-series of Recommendations

Best practices for implementing the circular economy in ICT management

1 Scope

This Supplement will contain a series of best practices for implementing the circular economy in ICT management. It will contain best practices of initiatives, policy instruments, circular processes, and other examples that aim to facilitate circular actions across the ICT value chain.

This work item also aims to provide impact analysis and, subsequently, devise general guidelines and goals for designing and implementing circular initiatives and policies for ICT management.

2 References

- [ITU-T L.1030] Recommendation ITU-T L.1030 (2018), *E-waste management framework for countries*.
- [ITU-T L.1031] Recommendation ITU-T L.1031 (2018), *Guideline on implementing the e-waste reduction target of the Connect 2020 Agenda*.
- [ITU-T L.1033] Recommendation ITU-T L.1033 (date), *Title*.
- [ITU-T L.1023] Recommendation ITU-T L.1023 (date), *Title*.
- [ITU-T L.1024] Recommendation ITU-T L.1024 (date), *Title*.
- [ITU-T L.1470] Recommendation ITU-T L.1470 (date), *Title*.

L.SM_Batteries?

3 Definitions

<Check in the ITU-T terms and definitions database at www.itu.int/go/terminology-database whether the term has already been defined in a Recommendation. It would be more consistent to refer to such a definition rather than to redefine the term>

3.1 Terms defined elsewhere

This Supplement uses the following terms defined elsewhere:

3.1.1 circular economy [ITU-T L.1020]: A circular economy is restorative and regenerative by design, and aims to keep products, components, and materials at their highest utility and value at all times while reducing waste streams.

A concept that distinguishes between technical and biological cycles, the circular economy is a continuous, positive development cycle. It preserves and enhances natural capital, optimises resource yields, and minimises system risks by managing finite stocks and renewable flows, while reducing waste streams.

NOTE – The definition is based on [b-EMF], and amended.

3.1.2 <Term 2> [Reference]: <optional quoted definition>.

3.2 Terms defined in this Supplement

<Note that terms defined in Supplements are not considered normative, and will not be added to the ITU terminology database >

This Supplement defines the following terms:

3.2.1 <Term 3>: <definition>.

4 Abbreviations and acronyms

This Supplement uses the following abbreviations and acronyms:

CE	Circular Economy
DfE	Design-for-Environment
EEE	Electrical and Electronic Equipment
EoL	End of Life
EPD	Environmental Product Declaration
EPR	Extended Producer Responsibility
ESM	Environmentally Sound Manner
HS	Harmonized System
ICT	Information Communication Technology IPR Individual Producer Responsibility
LCA	Life Cycle Assessment
NIE	Network Infrastructure Equipment
PRO	Producer Responsibility Organization
StEP	Solving the E-waste Problem
TBC	Take-Back Committee
TF	Treatment Facility
TPO	Third Party Organization
WEEE	Waste Electrical and Electronic Equipment

5 Conventions

None.

6 Commitments to circular ICT management

Global: SDG, ITU Connect 2030, NetZero, ...

Regional: e.g.:

EU: European Commission: Green Deal, Sustainable Product Initiative

Africa: African Circular Economy Alliance

Other regions

National: List ...

City and community: Barcelona City Council

7 Summary of circular ICT management use cases

Definition of all use cases, inspired and summarising existing cases of circular practice and policy mechanisms with known assessed impacts.

7.1 Digitalisation of circular design

Transparency in the design

Third party certification. Inspired by examples such as:

<https://tcocertified.com/iso-14024/>

ITU-T L.1023

A global digital sustainable product passport for ICT products. Examples:

<https://www.giplatform.org/sixth-dialogue-on-environmental-data-report/>

ITU-T L.GDSPP

7.2 Green and due diligence for public procurement of sustainable ICT products

Public procurement. Examples:

https://ec.europa.eu/environment/gpp/index_en.htm

<https://publications.jrc.ec.europa.eu/repository/handle/JRC124294>

Clauses, due diligence, circularity, recycling, SDG impacts, sustainability vs environmental impacts. Examples:

https://electronicswatch.org/en/delivering-due-diligence_2595038

Renewable energy initiative. Examples:-

As part of climate change efforts, one of India's telco viz. Reliance Jio Infocomm Limited has already set up over 144 MW of solar power at over 16,000 of its sites across India. These renewable energy efforts would feed into Telco and the parent Group's commitment to be Net Zero by the year 2035. Its emission reduction targets have also been validated by the Science Based Target Initiative (SBTi) of UNFCCC.

<https://www.aninews.in/news/business/reliance-jio-bags-esg-performance-in-telecom-sector-award20221123112943/>

In India, another telco viz. Airtel is reported to be using green energy to power its networks and data centres and has also reduced their carbon footprint to a large extent. It is committed to the Global System for Mobile Communications Association (GSMA) to transform the global mobile industry to reach net zero carbon emissions by 2050 by reducing absolute Scope 1 and 2 greenhouse gas (GHG) emissions by 50.2% till FY 2031 (from FY 2021 as the base year). In addition to this, Airtel has also committed to reducing absolute Scope 3 GHG emissions by 42% till FY 2031. To achieve these commitments following has been done by Airtel

a. Commissioned a photovoltaic asset with an installed capacity of 28MW via partnership with Amp Energy in Uttar Pradesh data centres

b. Sourced the electricity from a 9MW hydro power plant to its data centres in Delhi.

c. The group relies on sourcing ~6.5MW of renewable energy from wind and hydro assets in Karnataka and Tamil Nadu around the clock.

d. Partnered with **Avaada Energy** for sourcing solar power from 65MW of photovoltaic assets.

<https://www.gsma.com/betterfuture/wp-content/uploads/2022/05/Mobile-Net-Zero-State-of-the-Industry-on-Climate-Action-2022.pdf>

7.3 Repairability and repair

Repairability index. Example: France, etc.

<https://www.halteobsolescence.org/wp-content/uploads/2022/02/Rapport-indice-de-reparabilite.pdf>

Tax incentives for repair

<https://www.dinheirovivo.pt/economia/impostos/amp/reparacao-de-aparelhos-domesticos-e-queijos-vegan-passam-a-iva-de-6-14774258.html>

7.4 Circular business models

Servitisation (device as a service), e.g. home routers for Internet access

7.5 Reuse and refurbishment

... for digital inclusion. Example:

<https://circulartech.apc.org/books/a-guide-to-the-circular-economy-of-digital-devices/page/case-study-ereuse-building-reuse-circuits-for-social-inclusion>

Some companies working in ICT domain have started refurbishing and reselling their old equipment, such as mobile phones and routers. This reduces e-waste and extends the lifespan of the products, which is beneficial for both the environment and the company's bottom line.

Telecom companies have started reusing materials from their old equipment, such as copper and aluminum, in new products. This reduces the need to extract new materials, conserves resources, and reduces waste.

Vodafone aims to repair, resell and refurbish old electronics to eliminate waste and reduce CO2 emissions. In 2020, the company launched an internal asset marketplace, which allows the company's vast network of operators across different markets to repurpose excess stock and decommissioned hardware. This initiative is part of the company's goal to reuse, resell or recycle 100% of its network waste by 2025. To eliminate the need to supply plastic SIMs entirely, Vodafone launched Eco-SIM cards in October 2021. Out of the 8,800 tonnes of e-waste generated by Vodafone in 2022, the operator managed to recycle approximately 95% of its e-waste.

<https://www.mobiliseglobal.com/the-circular-economy-in-telecom/>

<https://www.mobiliseglobal.com/a-telcos-role-in-sustainable-business/>

7.6 Recycling

... for socio-economic and digital inclusion. Examples:

Formatted: Indent: First line: 1.27 cm

Formatted: Justified, Indent: First line: 1.27 cm

Formatted: Indent: First line: 1.27 cm

<https://circulartech.apc.org/books/a-guide-to-the-circular-economy-of-digital-devices/page/case-study-karo-sambhav-make-it-possible-working-with-manufacturers-to-create-an-e-waste-ecosystem-in-india>

<https://circulartech.apc.org/books/a-guide-to-the-circular-economy-of-digital-devices/page/case-study-the-microfactory-model-smart-innovation-for-urban-waste-mining>

It is reported that Nokia has adopted circular practices and made progress by setting clearly defined objectives. Their goal is to develop material management to such an extent that by 2030 there will be no need for landfilling. In 2020, Nokia processed 5,870 metric tons of obsolete products out of which 570 metric tons of products were reused. In the same year, Nokia put in motion a policy to improve the use of recycled materials in its products. 54% of the 23,200 metric tons of cast aluminium parts used in Nokia products had recycled materials in them. In 2020, Nokia tried to improve the recycling prospects of plastics. The company has also saved 31,600 tons of CO₂ through circular practices in that year

<https://www.nokia.com/about-us/newsroom/articles/embracing-circularity-for-a-sustainable-future/>

7.7 Public donation of decommissioned ICT devices

7.8 Extended producer responsibility

Extended producer responsibility (EPR) can be defined as the following:

A policy principle to promote total life cycle environmental improvements of product systems by extending the responsibility of the manufactures of the product to various parts of the entire life cycle of the product and especially to the take-back, recycling and final disposal of the product.

Benefits of EPR systems include: (from L.1031)

7.9 Environmental impacts: data transparency, accountability, verifiability

7.10 Communal involvement :- A sharing model for telecom tower sites can be an effective way to reduce costs and energy requirements for the industry. By allowing multiple service providers to host their active network components on a single tower site, companies can avoid the need to build redundant infrastructure and reduce the amount of energy required to power and cool equipment.

In addition, sharing tower sites can help to improve network coverage and quality, as it allows for better utilization of available resources. This can lead to a more efficient use of energy and resources, and ultimately result in a more sustainable and cost-effective business model for the industry.

7.11 Circular markets

https://ec.europa.eu/environment/strategy/circular-economy-action-plan_en

8 Catalogue of cases, initiatives and policy mechanisms

... how it could work, by example

Description of a few concrete examples with experiences of implementation that report results (benefits), related to the previous use cases.

8.1 Policy Mechanism: -

1. E-Waste Management Rules and Extended Producer Responsibility (EPR): A set of E-Waste Management Rules may be introduced mandating that only authorized dismantlers and recyclers will collect e-waste. It applies to every producer, consumer, or bulk consumer involved in the manufacture, sale, purchase, and processing of electrical and electronic equipment or components, as well as collection centre, dismantler, and recycler of e-waste. These rules hold the stakeholders responsible for complying with the mentioned guidelines and failure to do so results in severe penalties for non-compliant entities.

One of the cornerstones of E-Waste Management Rules is to provide Extended Producer Responsibility. EPR puts responsibility on any producer of electrical or electronic equipment for meeting recycling targets, that too only through registered recyclers of e-waste to ensure environmentally sound management of such waste. All producers are to fulfill their extended producer responsibility, and in doing so, they may also take the help of third-party organizations such as producer responsibility organizations, collection centres, dealers, etc. The producer shall fulfill their extended producer responsibility through the online purchase of extended producer responsibility certificates from registered recyclers only and submit them online by filing the quarterly return. The certificates shall be subject to environmental audit by an authorised agency.

Extended Producer Responsibility Framework:

1. **For Manufacturer:** To collect e-waste generated during the manufacture of any electrical and electronic equipment and ensure its recycling or disposal.
2. **For Producer:** To obtain and implement extended producer responsibility targets and create awareness through media, publications, advertisements, posters, or by any other means of communication.
3. **For Refurbisher:** To collect e-waste generated during the process of refurbishing and hand over the waste to the registered recycler and ensure that the refurbished equipment shall be as per the Compulsory Registration Scheme of Govt of India and standards of Bureau of Indian Standards (BIS) framed for this purpose.
4. **For Bulk Consumer:** To ensure that e-waste generated by them shall be handed over only to the registered producer, refurbisher, or recycler.
5. **For Recycler:**
 - Ensure that the facility and recycling processes are followed as per the standards or guidelines laid down by the CPCB in this regard from time to time
 - To ensure that the fractions or material not recycled in its facility is sent to the respective registered recycler

- To ensure that residue generated during the recycling process is not disposed of in an unauthorized treatment storage disposal facility
- To accept waste for recycling provided that they do not contain any radioactive material
- To create awareness through media, publications, advertisements, posters, or by such other means of communication
- To take the help of dismantlers for recycling purposes

6. For local Governments:

- To ensure recognition and registration of workers involved in dismantling and recycling
- To assist the formation of groups of such workers to facilitate the setting up of dismantling facilities
- To undertake industrial skill development activities for the workers involved in dismantling and recycling
- To undertake annual monitoring and to ensure the safety and health of workers involved in dismantling and recycling

2. There can be reforms like Know Your Customers (KYC) reforms, wherein self-KYC (App based) may be permitted. Further, Paper Customer Acquisition Forms (CAF) will be replaced by digital storage of data. These paper CAF may be used for printing of papers after pulping of these documents. These reforms have done away the environmentally straining paper based customer acquisition process.

9 Metrics for impact and implementation

Classification of different mechanisms by use case, considering pre-requisites or conditions for applicability and implementation, expected effect, obstacles, opportunities, scalability, contribution to circular commitments and the implementation of the circular economy.

Implementation of the circular economy:

Some action points identified to be incorporated into 'Circular Economy in Telecom sector'.

1. In view of the roll out of 5G services, a proposal is to prepare a Digital Readiness Index to facilitate monitoring mechanism which in turn will facilitate monitoring of Circular Economy Goals. Blockchain based approach can be used to monitor the whole process from acquisition of raw materials to its disposal.
2. Physical SIM cards may be substituted with e-SIM. Since a lot of unnecessary plastic is used in the manufacture of a SIM card, it is more beneficial to use an eSIM or embedded SIM. In this case, the SIM card will come attached to the phone thus eliminating the need for physical components. Consumers can install and activate the SIM through an app thus reducing paperwork and saving paper. This will reduce waste and customers too would not have to wait for the delivery of a SIM card.
3. Empirical study may be done to study of useful life of telecom products.
4. A comprehensive mapping of recycling facilities in India will be undertaken. Incentives needs to be provided for attracting recyclers and engagement with informal sector should also be increased.

5. To promote and use of Remanufactured and refurbished product, there is a need to have clear definition, standardisation/testing and certification.
6. **Cloud storage:** We use storage chips and devices like SD Cards, pen drives CDs, etc., for various information. But these products turn into e-wastes when disposed of. Sometimes they are hoarded even without proper use. Instead of using physical products like these, one can turn to cloud storage platforms for storing their information. It is not only easily accessible but it also gives more storage. By careful use of cloud storage, one will save space as well as will be able to reduce e-waste.
- 4.7.A Universal Charging Device (UCD) refers to a common charging port charging communications protocol across brands and devices with similar power requirements. A (UCD) will benefit users and the environment. Reducing the need for multiple charges reduces material use and e-waste generated, not to mention the headache and pocket. In India also, UCD is being explored for smartphones, tablets, e-readers, wearables. India's sale of smartphones alone has risen largely. Therefore, a drop in number of chargers will mean reduced demand of materials used to make these and thus will reduce waste.

10 Challenges in Implementation of the circular economy

10.1 CHALLENGES IN E-WASTE MANAGEMENT

Despite the clear benefits of circularity, transitioning into it from their traditional business models is not easy for the telecom industry. One of the complexities is related to the rate at which end-of-life products are generated, collected and shipped to a repair facility. The other aspects are related to consumers apprehension on data destruction. Economic viability of extraction of secondary raw materials and reuse may also pose difficulty. Overall, for the telecom industry to achieve its sustainability goals, tackling the generation of e-waste is crucial which includes greater lifespan to eco-friendly designs. This requires the telecom industry to be open to forging new partnerships with different players in the ecosystem as well as setting high goals for recycling, reusing, and championing the use of longer-lasting equipment.

10.2 HOARDING OF E-WASTE

Many people have a habit of what is called "Digital Hoarding", where they keep their old electronic products (functional and non-functional) instead of giving them away for recycling or donation. These can be for various reasons. It may seem to be harmless, but it can rather be harmful to the environment as it is happening on a large scale. Since the rare elements remaining inside these hoarded products are not used, more mining is done to retrieve them to make new products, thus resulting in the depletion of Earth's natural resources.

10.3 LACK OF TRUST

A lot of consumers do not recycle their telecom products, especially phones, SIM cards, storage chips, etc. fearing theft and misuse of data. They also avoid cloud storage as its not fully secure. Blancco Technology Group concluded that India roughly spends more than 2 million dollars annually for destroying data storage devices . Making people aware of alternate storage sources and their benefits will help reduce e-waste.

10.4 Lack of harmony between the stakeholders

It is not simply the responsibility of telecom companies (telcos) to drive a circular economy. While telcos need to promote products that are made from recycled materials and have a greater lifespan to reduce waste, vendors too will need to provide products that meet this requirement. Therefore, the industry needs to be open to new partnerships to progress in circular economy efforts and harmony between the players in the ecosystem. Sustainability is becoming a strategic goal for most industries globally, and India's telecom industry needs to catch up.

Appendix I

<Appendix title>

<Note that it is preferable not to have annexes in Supplements, only appendices. The use of annexes is in practice reserved for Recommendations. Include here the content of appendix I>

Bibliography

<Optional – As Supplements are non-normative, all references can be placed in clause 2. However, some authors prefer to also list additional documents in a bibliography>

[b-ITU-T X.yyy] Recommendation ITU-T X.yyy (date), *Title*.
